Heat and Humidity

- Mines air-conditioning employs the processes like
  - Cooling and
  - Dehumidification.
- Therefore, design of mine air-conditioning systems require estimating the heat generated by different sources in underground mines.

SOURCES OF HEAT IN MINES

The potential sources of heat in mines are

Heat is emitted into subsurface ventilation system from a variety of sources.

- Surface air entering the mine
- Heat flow from exposed wall rock
- Heat due to auto-compression in shafts and near-vertical openings
- Ground water
- Machinery and lights, locomotives
- Human metabolism
- Oxidation
- Blasting
- Rock movement

High temperature of surface air entering the mine

- Surface air entering the mine carry with it some heat if the surface temperature is high as compared to the underground temperature.

Heat due to auto-compression in shafts and near-vertical openings

- Temperature of air changes with change in pressure.
- Compression of air increases air temperature and vice versa.
- Due to auto compression, potential energy in converted to thermal energy.
- As the air descends the downcast shaft, it gets compressed by the weight of the shaft air-column approx. at a rate of 1.1 kPa per 100 m depth and its potential energy is converted to heat energy.
- Due to rise in pressure, the temperature of the compressed air may rise 5 to 7 times than the ventilation air.
- There could be a transfer of heat from the compressed air line to the ventilation air.

- If no exchange (loss or gain) in the heat or moisture content of the air takes place in the shaft, the compression occurs adiabatically, with the temperature rise following the adiabatic law.

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

Where

T = temperature in K

γ = Cp/Cv = 1.404 for dry air (it varies slightly with the moisture content of air, but for mining purposes it can be taken as 1.4)

V = specific volume (volume of unit mass) of air

P = barometric pressure and

Subscripts 1 and 2 indicate the state of air at the shaft-top and shaft-bottom respectively.

- Assuming compression to be adiabatic, the rise in temp. due to auto compression can also be obtained by equating the potential energy with enthalpy change

$$dQ - dW = dH + dPE + dKE$$

Where

dQ = heat added to or removed from the section

dW = external work done on or by the air in the section

dH = change in enthalpy of the air across the section

dPE = change in potential energy of the air across the section = *gdh*

dKE = change in kinetic energy of the air across the section

Under the assumptions made,

As no heat is transferred, dQ = 0

As no work is done, dW = 0

As the flow is non-accelerative, dKE = 0

So that **dH = - dPE and hg = ΔH = CpΔT**

Where

ΔT = rise in temperature, K

ΔH = rise in enthalpy, J/kg

h = depth of shaft, m

Cp = specific heat of air in J/kgK

$$H = mgh/1000$$

Where

m = mass of air, kg

g = acceleration due to gravity, m/s^2

h = depth of shaft, m

H = increase in heat content, kJ/kg

- The increase in heat due to auto compression of 1 kg of air passing 100 m vertically downcast shaft is given by:

$$H = (9.8 \times 1 \times 100)/1000 = 0.98 \text{ kJ/kg}$$

Heat from rock

- Heat flow from underground wall rock is a major contributing source to the rise in mine air temp.

- Amount of heat transfer from the rock, in a given mining condition, requires the knowledge of thermal properties of rock.

- Heat flows out of the hot core of the earth at almost a constant rate of 0.05 W/m^2 over most of the earth's surface.

- As a result, the temp. of the ground rock increases steadily as we go deeper down the earth's crust.

- The rate of increase of temp. with depth is called **geothermal gradient**.

$$\text{The geothermal gradient, gg} = \frac{0.05\,(\text{w/m}^2)}{\text{thermal conductivity (w/m.}^{\text{o}}C)}\,({}^{o}C/m)$$

- Geothermal gradient varies from place to place and it dependent on the following physical properties of the rock which govern the rate of heat transfer in the rock as
  - Thermal conductivity: In rocks of lower thermal conductivity such as coal-measure rocks, the geothermal gradient is steeper or the rise in temp. with depth is faster.
  - Specific heat and
  - Density.

- The typical values of geothermal gradient (°C/100m) are as follows:

Ontario, Canada	1.22
Hungary	5.00
KGF, India	1.10
Wit Waterstrand, South Africa	0.80
UK and Europe	1.00 – 3.00
Broken hill, NSW, Australia	1.97

Virgin rock temperature (VRT)

- In most climates the ground temp. at a depth of 15 m is not affected by changes in surface air temp, and this ground temp. is called **virgin rock temperature**.

- This temp. which varies from place to place depending on the climate, remains constant throughout the year and is usually the average annual temp. at the surface.

- The geothermic gradient at any place is thus the rate of rise in temp. above this temp.

- Direct heat transfer from the rock to mine air is governed by the rate of heat transfer within the rock mass and the heat transfer from rock wall of the excavation to the mine air.

- Heat transfer from the rock to mine air is mainly through direct heat transfer from the exposed rock surface to the air.

- When rock surface is dry, the heat transfer is mainly through convection and raises the sensible heat of the air, but when the rock surface is wet a substantial amount of water evaporates into the air causing both sensible and latent heat transfer.

- The rate of sensible heat transfer from rock wall to the air is a linear function of the difference between the temperature of the rock wall and the mean air temperature as well as the coefficient of heat transfer

$$q = \alpha (Ts - Ta)$$

Where

q = rate of heat transfer / amount of hat flow per unit area per unit time

α = coefficient of heat transfer and

Ts and Ta are the rock-wall and mean air temperatures respectively.

Heat transfer from wet rock surface

The moisture evaporated from the surface of airway considerably increases the rate of heat transfer by extracting the latent heat of evaporation from the rock. The combined sensible and latent heat transfer in wet airways is given by

$$q = \alpha (T_s - T) + f_w \, l \, \varepsilon \{e'(T_s) - e\}$$

where T = dry-bulb temperature of air

f_w = fraction of airway surface that is wet

l = latent heat of evaporation of water

ε = coefficient of mass transfer

$e'(T_s)$ = saturation vapour pressure at T_s, which is the overall surface temperature for both dry and wet portions of the rock = $aT_s^2 + bT_s + c$

where a, b and c are constants having values of 0.002133, -1.12164 and 147.83 respectively with Ts in K and e' in kPa

e = vapour pressure of air = 461.9×10^{-6} m' T kPa, where m' is absolute humidity, g/m^3

Ground water

- All ground water, especially from hot fissures and natural rock reservoirs, is a prolific source of heat in mine workings.

- Since the water and heat both are derived from the surrounding rock or geothermic sources, the water temperature will approach or even exceed that of the rock.

- The strata water oozes out almost at the virgin-rock temperature and cools rapidly by evaporation as soon as it is exposed to the mine air.

- During evaporation, the water transfers its heat to the mine air increasing the latent heat of air.

- **Heat transfer through strata water = Cpw (Tv - T) per unit mass of water flowing out.**

Where, Cpw = sp. heat of water, Tv = virgin rock temp. and T = temp. of water entering the sump

- A survey of 7 hot mines in US and Canada showed that mine water added 20% of the total heat gained by the air.

Heat from man

- Heat is produced by men through the process of metabolism.
- Even a man at rest produces quite an appreciable quantity of heat by basal metabolism (when food is withheld for a specific length of time).
- It is estimated that the heat produced by basal metabolism is 46.5 W/m^2 of body surface.
- Average men have a body surface of 1.8-1.9 m^2 and have a basal metabolic heat production of 84-88 W.
- When doing hard work, the heat produced by the body is much more and sometimes as much as 10 times than that produced by basal metabolism.
- A typical average heat-generation rate thorough metabolism is around 200 or 300 W per person.
- However, hard-working men can generate metabolic heat up to 400 or 500 W for short periods of time.

Heat produced by machinery

- In highly mechanized mines, this can be a large source of heat as all the energy consumption of u/g machinery adds heat to the mine air.

- The power losses and most of the work done are converted directly to heat or indirectly to heat through friction.

- Almost all the work done by face machinery for cutting, drilling, loading and transport is frictional and most of the power input to such machinery is converted to heat partly in the machine itself and partly through frictional work.

- Therefore, amount of heat generated by electric-powered machinery in a mine appears to equivalent to the electric power input.

- Most of the heat produced by the face machinery increases the temp. of air at the face.

Heat from lights

- Heat produced by different light sources depends on the current and voltage.

Light sources	**Heat production**
A candle	25-35 J/s
An electric cap lamp (two cell type)	2.6 J/s
An ordinary electric bulb	40 J/s

- Use of carbide lamps produces a significant amount of heat.

- It is estimated that a carbide lamp consuming 156 g of carbide per shift produces 96.5 W of heat and it may be as high as 204.7 W when it burns at full brightness.

Heat due to oxidation

- Oxidation process involving the mineral, backfill, and timber in mines contribute heat to the mine air.

- This is a major source of heat in coal mines, particularly in seams liable to spontaneous heating.

- In coal mines, 80-85% of the heat added to the air can be traced to this source.

- Heat due to oxidation is not appreciably high in metal mines where a small quantity of heat may be produced by the oxidation of timber.

- In ore mines producing sulfide ores, oxidation of sulphides may add a considerable amount of heat to the air.

- Heat due to oxidation of coal is 8.79 MJ/m^3 of O2 absorbed while that for oxidation of pyrites is 18 MJ/m^3.

Heat due to blasting

- Blasting is a significant heat source and can be of considerable magnitude.
- Over half and perhaps 90% of the energy released by the detonation of high explosive is liberated in the form of heat.
- The amount of heat released depends on the type of explosive used in blasting.
- Heat released varies from about 3700 kJ/kg for ANFO to 5800 kJ/kg for nitroglycerine.
- It is estimated that heat produced by blasting in a mine milling 101 600 tones per month to be 316.5 MJ/h on an average, but the actual heat produced in the hour of blast is of order of 5.3-6.3 GJ.
- However, this heat is dissipated away by the ventilating air current before men return to work after blasting.

Heat caused by rock movement

Sources of heat due to rock movement:

- Movement of ground due to geologic causes or mining subsidence
- Caving or collapse of waste or ore in stopes or abandoned areas: it is the most common cause of heat liberation due to ground movement.

- However, the actual heat addition to the air due to movement of strata in coal mines is only 1 % of the total heat added to the air. But theoretically it has been found to be around 9%.
- This is believed to be due to most of the heat being dissipated in the broken rock mass itself.

PHYSIOLOGICAL EFFECTS
OF
HEAT AND HUMIDITY
ON THE MINER

- ‘Mine climate’ refers to the prevailing conditions of
  - temperature,
  - humidity and
  - velocity of air in a mine.
- In deep mines, intensively mechanized mines and working areas farther from shafts, climatic conditions deteriorate due to increase in temperature and humidity, influencing the health, safety and efficiency of the miners.
- As temperature and humidity increases,
  - accident rate increases and
  - working efficiency decreases and hence
  - production decreases.

- Human body employs a remarkable control system for regulating body heat and holding temp. nearly constant at 98.6 °F (37 °C).
- This mechanism maintain a balance between the heat-loss and heat-gain to prevent harmful thermal effects (heat strain) to the body.
- Metabolism is the source of all heat produced within the body.
- At rest in a comfortable environment, due to oxidation and other chemical processes while digesting food, a person liberates heat of about 115 W, is termed **basal metabolism**.
- In addition, 115 to 585 W or more of waste heat is produced during physical exertion.
- Waste heat produced by the process of metabolism is dissipated through skin into the surrounding mine air by conventional heat-transfer processes:
  - Convection,
  - Radiation and/or
  - Evaporation of sweat.

- Small part of heat is given out through exhaled air during respiration.

<u>Heat-balance equation of human body</u>

- A basic heat-balance equation is developed by the **American Society of Heating, Refrigerating & Air-conditioning Engineers (ASHRAE)**, to express the heat changes in the human body.

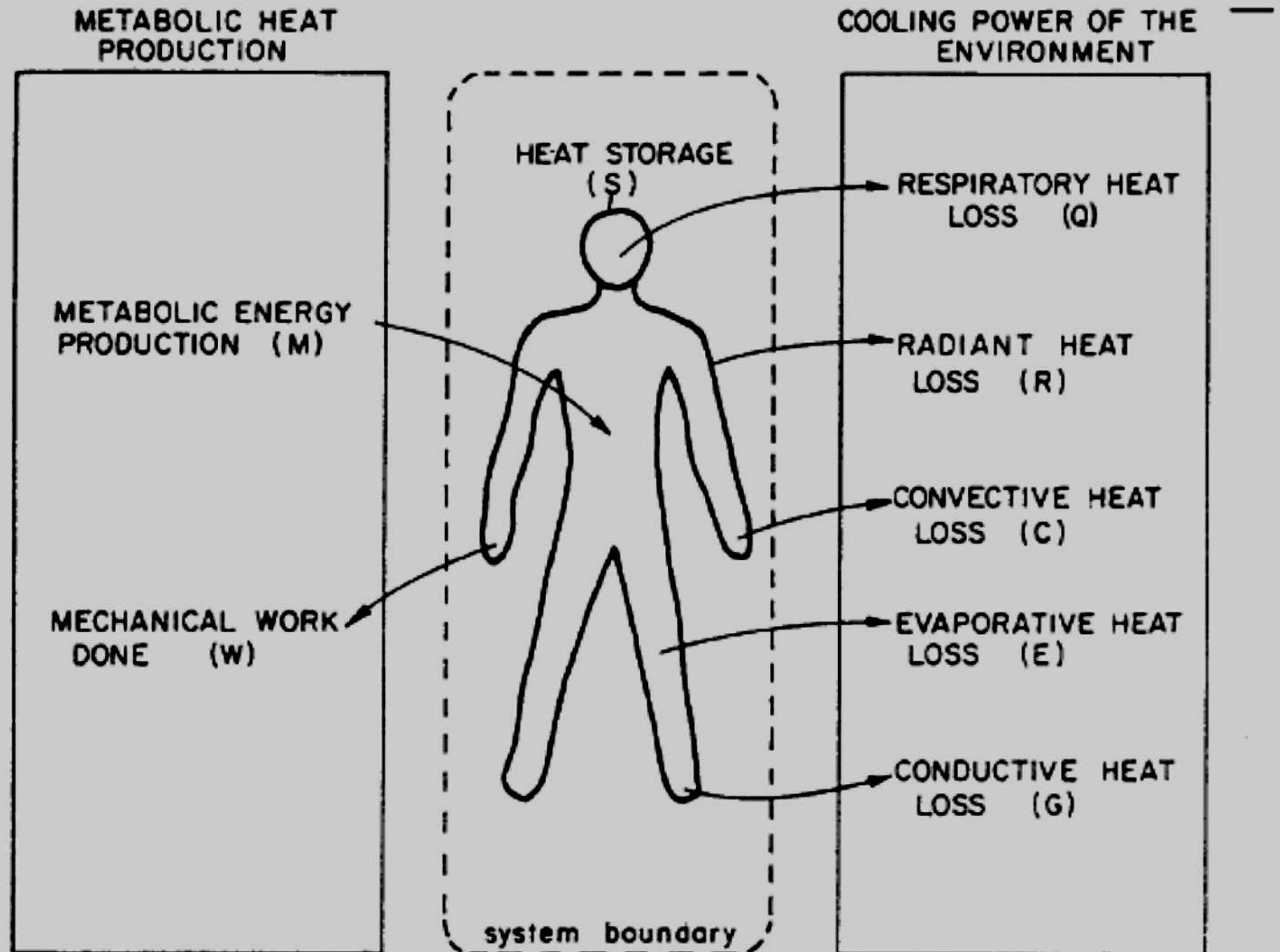
- Heat losses from the body are considered +ve and heat gains are –ve.

- Since mechanical work (e.g. climbing a ladder) is accomplished by the body, it is taken as +ve and hence subtracted from metabolism to find the net body-heat production.

- Work is –ve when potential energy is added, e. g. walking down steps.

- Symbolically, heat balance for the human body can be represented as in fig.

The Heat Balance of a Working Person[2]



Schematic representation of an energy balance on the human body.

To understand human heat strain the use of the heat balance equation is important:

$$M - W - Q - S = R + C + E + G$$

This equation provides the basis for quantifying heat stress and relating human physiological response to the heat stress causing it.

Effects and symptoms of heat illness/stress on miners

Heat stroke:

- It is the most serious of heat disorders and 20 to 80% are fatal.
- Due to failure of heat regulatory mechanism of the body
- At body temp. more than 40 °C, sweating stops, leaving the skin hot, dry, and flushed.
- The patient may sink into a coma.

Heat cramps:

- They are the acute form of salt depletion.
- Especially common among workers in hot factories and mines.
- Symptoms: fatigue, dizziness and severe muscle pain, leading to stomach cramp.

Heat exhaustion:

- Less severe but chronic.

- Marked by fatigue, headaches, dizziness, blurred vision, and sometimes an inability to sweat.

Mental fatigue:

- Due to continued undersupply of blood to brain.

- Symptoms are Carelessness, Rebellious attitude, Neglect to work

Other heat illnesses:

- Heat fainting, the most common.

- Heat retention

- Dehydration